

Assorted Notes

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1 Six Operations

A good source is Gallauer 2022.

One important example is the relative point of view as discussed in Gallauer 2022. One discovers invariants that are easy to compute. An example would be rational points of a variety over a finite field. Then, one finds a corresponding cohomology group associated with this. In this case, it is the l -adic cohomology. This idea of a cohomology group was explored by Galois. In André 2008, it is ambiguous what you want to consider given a constant invariant, and this ambiguity is captured by a group.

To find a higher structure of cohomology, one can consider a cohomology group to be something local. To assemble something local to something global, one invents the idea of a l -adic constructible

derived category. This is a derived category of constructible sheaves. Sheaves, or higher versions of sheaves with better combinatorics, give the right properties.

To invent the notion of global to local correspondences, one needs some form of restriction on coverings. Hence, one invents the notion of a presheaf as a contravariant functor of space, most a combinatorial model such a simplicial set. Then, this is valued in an $(\infty, 1)$ -groupoid to capture automorphisms. nLab authors 2023 gives a definition.

The more basic example is a presheaf as a contravariant functor from the category of open sets of a topological space to the category of sets. One loses the ability to capture automorphisms. Set valued presheaves cannot capture automorphisms. For sheafification, one wants separability or the representatives being unique on overlaps, and descent, which is a condition for covering sieves. For descent, or gluing, Čech cocycles are effective with an equaliser diagram. The purpose of descent is to capture the idea of a local to global correspondence.

We now work in the category of morphisms of schemes. Why should I care about adjoint functors? This morphism induces adjoint functors which are ordinary pullback and ordinary pushforward between categories of sheaves.

2 Algebraic geometry

2.1 Higher category theory and derived algebraic geometry

A good reference is Scholze 2026.

I also intend to read Aoki's work on Higher Zariski Geometry. There is an adjunction between commutative algebra objects to the opposite category of locally ringed spaces. The spectrum functor assigns to a commutative its Zariski spectrum with a structure sheaf.

The duality between affine schemes and commutative rings associating affine schemes to regular functions. One has to restrict to affine objects on the geometric side and glue them.

This is because affine objects have the nice property endowed by definition that higher direct images vanish or that affine morphisms are those where pullbacks are t -exact. Recall that pullbacks are motivated by thinking about cohomology in topological space, and then trying to find adjoint functors associated with morphisms of schemes.

One then, using Serre's affineness criterion to define affine morphisms in more general contexts as those whose higher direct images vanish since the pullback is t -exact.

Consequently, one uses the basic idea of affine objects where for affine opens the inverse image is affine as a foundation. Then prove Serre's affineness criterion, then prove using derived algebraic geometry that the pullback is t -exact. I have not seen a good solution to this problem in the literature or a more general definition of affineness. It is hoped that Scholze 2026 side steps these issues with the construct of Stefanich rings and the gestalten to recover perfect duality between algebra and geometry.

A t -structure is therefore motivated by wanting objects in the heart to form an abelian category. Then, one can define exact functors that maps the heart of the first t -structure to the heart of the second t -structure. One can then define a t -exact functor as one that is exact on the hearts and hence ask for the pullback to be t -exact.

Separability means diagonal pullback is a closed immersion.

One invents the notion of a diagonal map to understand separability. The diagonal map is the prototypical example of self intersection in a space.

2.2 Computation on cohomology of projective space and line bundles

This follows Chapter 18 of Vakil. I also saw a calculation done by Akhil Mathew using Koszul cohomology on his blog. Other sources might be.

Eventually one wants to compute cohomology of line bundles in projective space as the first example of using Serre’s vanishing theorem in practice, seeing Serre duality for the first time.

I also want to eventually compute or derive the Riemann-Roch formula.

3 Stacks

4 Higher category theory

5 Chromatic homotopy theory

6 Equivariant homotopy theory

7 K-theory

I want to talk about using trace maps, then TCC and THH to assemble algebraic K-theory.

8 Mathematical physics

Good sources for this is D. B. Skinner’s Advanced Quantum Field Theory course notes.

9 Prismatic F-gauges

Jacob Lurie’s introductory talk is most accessible for this.

References

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